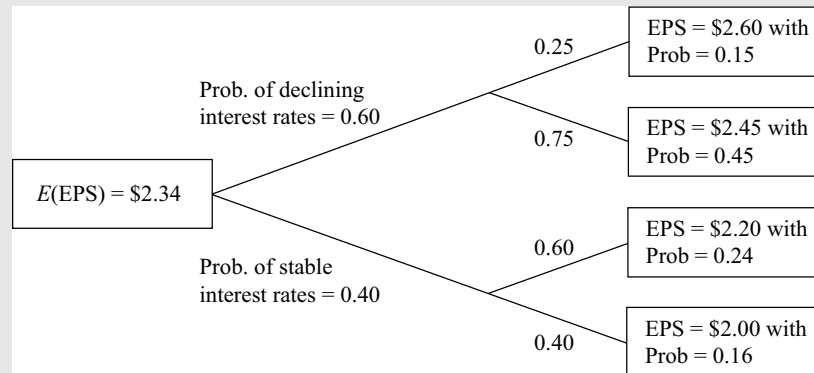


Exhibit 2: BankCorp's Forecasted EPS

A declining interest rate environment points us to the **node** of the tree that branches off into outcomes of USD2.60 and USD2.45. We can find expected EPS given a declining interest rate environment as follows, using Equation 6:

$$\begin{aligned} E(\text{EPS} \mid \text{declining interest rate environment}) \\ &= 0.25(\text{USD}2.60) + 0.75(\text{USD}2.45) \\ &= \text{USD}2.4875 \end{aligned}$$

If interest rates are stable,

$$\begin{aligned} E(\text{EPS} \mid \text{stable interest rate environment}) &= 0.60(\text{USD}2.20) + 0.40(\text{USD}2.00) \\ &= \text{USD}2.12 \end{aligned}$$

Once we have the new piece of information that interest rates are stable, for example, we revise our original expectation of EPS from USD2.34 downward to USD2.12. Now using the total probability rule for expected value,

$$\begin{aligned} E(\text{EPS}) \\ &= E(\text{EPS} \mid \text{declining interest rate environment})P(\text{declining interest rate environment}) \\ &\quad + E(\text{EPS} \mid \text{stable interest rate environment})P(\text{stable interest rate environment}) \end{aligned}$$

So, $E(\text{EPS}) = \text{USD}2.4875(0.60) + \text{USD}2.12(0.40) = \text{USD}2.3405$ or about USD2.34.

This amount is identical to the estimate of the expected value of EPS calculated directly from the probability distribution in Example 1. Just as our probabilities must be consistent, so too must our expected values, unconditional and conditional, be consistent; otherwise, our investment actions may create profit opportunities for other investors at our expense.

To review, we first developed the factors or scenarios that influence the outcome of the event of interest. After assigning probabilities to these scenarios, we formed expectations conditioned on the different scenarios. Then we worked backward to formulate an expected value as of today. In the problem just worked, EPS was the event of interest, and the interest rate environment was the factor influencing EPS.

We can also calculate the variance of EPS given each scenario:

$$\begin{aligned} \sigma^2(\text{EPS} \mid \text{declining interest rate environment}) \\ &= P(\text{USD}2.60 \mid \text{declining interest rate environment}) \\ &\quad \times [\text{USD}2.60 - E(\text{EPS} \mid \text{declining interest rate environment})]^2 \\ &\quad + P(\text{USD}2.45 \mid \text{declining interest rate environment}) \end{aligned}$$